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Mitigating Heterogeneity-Induced Acid Overflush in Reservoir: Smart Flow Control Completion and High-Resolution Numerical Modeling for Enhanced Acidizing

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Abstract

To address the challenges of heterogeneity-induced acid overflush in carbonate reservoirs, a Smart Flow Control Completion is developed to solve the problem of uneven acidizing effects in long horizontal wells. To better design the Smart Liner(SL), it is necessary to study the near-wellbore flow behavior of SL. In this paper, we develop an SL flow prediction model using COMSOL-based numerical simulations. Using this model, we simulate wormhole formation in carbonate reservoirs with varying porosities via a two-scale approach. The results demonstrate that adjusting hole density based on porosity variations—reducing holes in high-permeability zones and increasing them in low-permeability regions—improves stimulation uniformity. Results show the SL system balances flow distribution, mitigating overflush and enhancing wormhole development.

Introduction

Carbonate reservoirs hold approximately 60% of the world's oil reserves and are very important reservoirs [1]. Acidizing is one of the commonly used methods to increase production. Currently, the development of low-permeability reservoirs is gradually becoming a focus in the oil industry. Low-permeability reservoirs are characterized by poor porosity and permeability, strong reservoir heterogeneity, and poor water injectivity, which pose high technical challenges during the development process. To develop low-permeability reservoirs, techniques such as fracturing and acidizing are required to enhance oil well productivity [2].

During acidizing operations in horizontal wells, two major challenges significantly impact treatment efficiency: heel-to-toe effect and reservoir heterogeneity. The heel-to-toe effect arises from frictional pressure drop along the wellbore, causing uneven acid distribution with excessive fluid entry near the heel and insufficient coverage at the toe. Meanwhile, reservoir heterogeneity—variations in porosity, permeability, and mineral composition—leads to preferential acid flow into high-permeability zones,

bypassing low-permeability regions. These issues result in acid overflush, where excessive acid volume is wasted in already highly conductive zones, while under-stimulated areas remain poorly treated. It not only reduces stimulation effectiveness but also increases operational costs. To address these challenges, we propose a Smart Flow Control completion (Smart Liner, SL) system that automatically calculates wormhole PV numbers for each heterogeneous formation segment based on reservoir parameters (porosity and permeability) from the database. The system then converts PV numbers to equivalent wormhole lengths and skin factors, and accordingly adjusts the hole configuration to generate an optimized perforation scheme for uniform stimulation

As shown in [figure 1](#), the Smart Liner (SL) is a device used to control the distribution of acid during acidizing. Its basic principle is to control the discharge flow of acid by unevenly distributing the orifices (holes), achieving a uniform discharge of acid overall. The configuration and quantity of the pre-drilled holes were determined by pressure difference along the wellbore. This arrangement balances the uneven flow caused by the uneven pressure difference. SL does not have a complex mechanical structure. Instead, it passively balances the flow by changing the density, position, and orientation of the holes, utilizing the throttling effect of the holes.

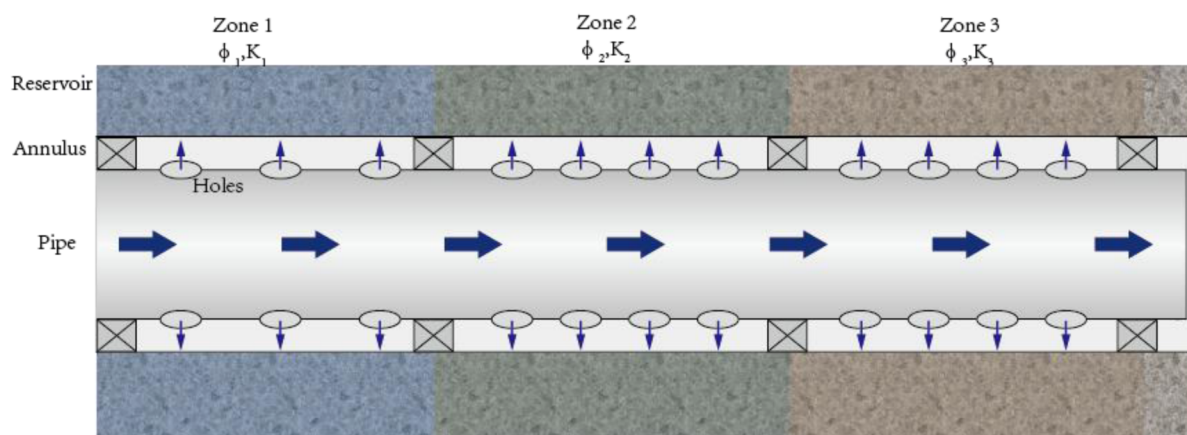


Figure 1—Smart Liner in heterogeneous formations (Using variable perforation density across different formation zones)

To improve the effectiveness of acidizing, it is necessary to reasonably adjust the distribution of holes in the SL [3–5]. To do this, it is first necessary to predict the flow distribution of the SL and the reservoir of acid-etched wormholes. The flow near the wellbore of SL involves acid-rock chemical reactions and multiphase flow in porous media, which requires focused research.

To investigate the near-wellbore flow characteristics of SL, it is necessary to first understand the pressure and flow conditions of SL. In current research, Kristian Mogensen et al. have proposed a relatively good model that is based on the continuity of pressure and flow. This model uses discrete nodes for iteration and matching, and automatically adjusts the number of perforations to ensure uniform flow in each section of the SL. In terms of predicting the pressure and flow of SL, this paper adopts the same approach as Kristian Mogensen, with simplifications and improvements made on this basis. The focus of this paper is on the near-wellbore jet flow through perforations; hence, instead of using artificially placed nodes, the corresponding perforation position nodes are directly used for calculations. This also facilitates the free adjustment of perforation locations. Kristian Mogensen's model employs a semi-empirical formula by M. Buijse et al. to predict wormholes and uses empirical parameters provided by Talbot et al. This approach is relatively crude. In reality, as the formation's physical properties change, the related parameters will also vary. While Mogensen's model employs fixed empirical values to predict acidizing outcomes, in heterogeneous formations the semi-empirical parameters (particularly PVBT) used in the model vary significantly rather than remaining constant. This variation causes actual wormhole lengths to deviate from

predicted results - being either longer or shorter than expected - leading to miscalculations in acid overflush requirements. In the area of acid etching simulation, there are many relevant studies. The dual-scale model proposed by Panga et al. effectively reflects the process of acid jet interaction with rock to form wormholes and has been widely adopted by many researchers. Therefore, this study is mainly based on the theory of Panga et al., implemented in COMSOL, and combined with the SL flow prediction program to simulate the acid etched wormholes formed by SL. This further allows for the evaluation of the acid etching effect of SL.

This paper aims to establish an internal flow prediction program for SL, with mathematical modeling and numerical simulation. By integrating the simulation of acid-etched wormhole reservoir in carbonate reservoirs, the production status of SL was evaluated, thereby enhancing the production capacity through SL acidization. The paper approaches the topic from two angles: SL flow prediction and the growth of acid-etched wormholes. Based on real data, predictions and assessments were made for two case studies, and design adjustments based on wormhole length were proposed.

SL Flow Prediction Model

Establishment of mathematical models

To determine the jet flow rate of each hole in the SL, it is first necessary to obtain velocity and pressure data inside the SL pipe. To achieve this goal, a mathematical model has been established, which focuses on calculating the resistance conditions of the pipe. The resistance of the pipe is mainly divided into two categories: the frictional resistance of the acid flowing through the pipe, and the local resistance when the acid flows out of the holes. The frictional resistance of the pipe can be calculated by eq. (1):

$$\Delta P_f = \lambda \cdot \frac{L}{D} \cdot \frac{\rho v^2}{2} \quad (1)$$

with L the length of the pipe, D the diameter of the pipe, ρ the density of the acid, v the velocity of the acid, λ the friction factor.

Since the flow within the pipe is essentially turbulent, the calculation of ' λ ' relies on empirical formulas. Here, the calculation is performed using the eqs. (2)[6,7]:

$$\frac{1}{\sqrt{f}} = -4 \log_{10} \left[\frac{1.26}{Re \sqrt{f}} + \frac{3.7\varepsilon}{D} \right] \quad (2)$$

with Re the Reynolds number, ε/D the relative roughness. In this paper, ' ε/D ' is taken as 0.0001. f is called Fanning friction factor, and $\lambda=4f$.

The pressure drop of hole depends on[8]:

$$\Delta p_{\text{hole}} = 2.23 \times 10^5 \times \frac{q_{\text{hole}}^2 \rho}{D_{\text{hole}}^4 C_D^2} \quad (3)$$

with ρ the acid density, D_{hole} the diameter of the hole, q_{hole} the flow rate through the hole, C_D the hole friction factor, C_D depends on hole size and diameter, in this paper, we take as 0.73.

Calculation process

Considering that the flow rate in the SL pipe is constantly changing, the pipe's friction coefficient and the pressure drop across the holes are also constantly changing. The velocity and pressure parameters of each hole are related to all the previous holes. In order to obtain flow data for each hole, our mathematical model divides the entire pipe into several basic units for calculation.

There are two types of basic units, corresponding to the two types of pipes in the actual SL. The segments between holes are called blind pipe units. The characteristic of the blind pipe segment is that there is no flow diversion through the holes, meaning the pipe flows at a constant rate and is only governed by the

friction loss formula. Blind pipe units only experience pressure changes, not flow changes. The resistance coefficient of the blind pipe units is obtained through the Newton-Raphson method.

Near the holes, the units are referred to as hole units. These units have no length and are only governed by the local resistance formula. The hole units only experience changes in flow rate, not in pressure. In fact, there is still frictional resistance near the holes, but since these units represent only the point at the hole, we consider that there is no change in frictional resistance in this vicinity. The treatment of the hole units is an idealized approximation and does not fully reflect the flow in the tee section near the holes, but it is sufficient for predicting the flow rate at the holes.

After clarifying the governing equations for the two types of units, the workflow of the model is as follows: First, the pipe is divided according to the distribution of holes in the SL, in a manner of hole unit - blind pipe unit - hole unit. Then, parameters such as the inlet flow rate are input. The model initially determines the pressure, calculates the pipe pressure and flow rate, and passes these values downstream. The pressure is repeatedly adjusted until the flow rate meets the requirements.

Figure 2 illustrates the general process of the aforementioned calculation method. The system initially presets a small pressure P_0 . Subsequently, it performs pressure and flow calculations based on the corresponding unit arrangement. Both pressure and flow follow continuous formulas, such as $Q_2 = Q_1 + \Delta Q$ and $P_1 = P_0 + \Delta P$. After calculating the flow rate at the inlet, the program compares the result with the preset Q_{inlet} . If it does not match, the system will continue to increase the pressure.

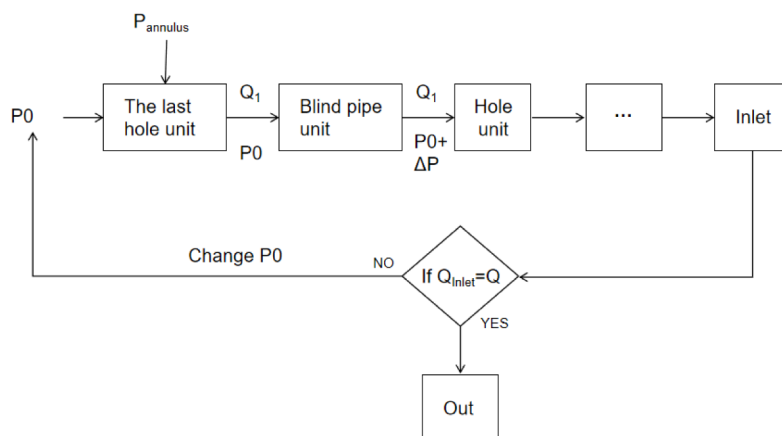


Figure 2—The calculation process

Wormhole Growth Simulation

As mentioned earlier, the acid flows through the pipe and eventually jets out from a certain hole. After the acid jets out, it passes through the annular space and comes into contact with the wellbore reservoir, where the hydrochloric acid that comes into contact with the carbonate rock quickly reacts with it, forming a certain cavity or fracture, known as a wormhole. Wormhole is an important reason for the increased production from acidizing techniques, and therefore their related parameters are the basis for evaluating the effectiveness of SL modifications. Generally, we are most concerned with the size, or length, of the wormholes. The development of wormholes involves multiple physical and chemical fields, including acid-rock reactions, multiphase flow in porous media, and changes in rock porosity and permeability. To study the length of wormholes formed by SL, we used COMSOL software for numerical simulation.

To simulate the wormhole growth, we modeled based on the theoretical model proposed by Panga[9]. The two-scale continuum model is one that operates at a scale between the core and pore scales, referred to as the Darcy scale. It is based on Darcy's law and chemical kinetics to control the flow of fluids, while introducing parameters such as porosity and pore diameter at the pore scale to control the Darcy scale model.

This indirectly reflects the influence of the pore structure while avoiding the need to establish a pore-scale model based on actual core samples.

Darcy Scale

The flow of acid in porous media is based on Darcy's law:

$$v = -\frac{K}{\mu} \cdot \nabla P \quad (4)$$

with K the permeability, μ the viscosity of the acid, ∇P the pressure gradient.

The rate of acid chemical reaction is defined as:

$$R(C_s) = k_c(C_s - C_f) \quad (5)$$

With k_c the mass transfer coefficient, C_s the concentration of acid at the solid-fluid interface, C_f the cup-mixing concentration of acid in the fluid phase.

Porosity consumed by the reaction can be expressed as:

$$\frac{\partial \phi}{\partial t} = \frac{R(C_s)a_v \alpha}{\rho_s} \quad (6)$$

with ϕ the porosity, a_v the interfacial area per unit volume of the medium, α the dissolving power of acid, ρ_s the rock density.

Pore Scale

The progress of chemical reactions leads to changes in porosity, which in turn affects the changes in permeability, pore size, and specific surface area :

$$\frac{K}{K_0} = \frac{\phi}{\phi_0} \left(\frac{\phi(1-\phi_0)}{\phi_0(1-\phi)} \right)^{2\beta} \quad (7)$$

$$\frac{r_p}{r_0} = \sqrt{\frac{K\phi_0}{K_0\phi}} \quad (8)$$

$$\frac{a_v}{a_0} = \frac{\phi r_0}{\phi_0 r_p} \quad (9)$$

with r_p the pore radius, and r_0 , K_0 , ϕ_0 , a_0 refer to the initial value of the corresponding parameters

Based on the above formulas, a coupled model of chemical field-seepage field-pore field can be established in COMSOL software. After setting the inlet velocity, acid concentration, and porosity, the wormhole growth condition can be simulated.

Design Example

After setting up the numerical model, we solved it based on the operating parameters of a certain horizontal well. Using the flow prediction model from earlier, we solved for the SL flow distribution in two scenarios: uniform distribution of holes and a gradient from sparse to dense holes from the heel to the toe of the well.

The relevant parameters for the example are as follows:

Table 1—The value of example

Wellbore diameter/mm	152.4
SL inner diameter/mm	100.58
hole diameter/mm	3
Inlet rate/(m ³ /min)	1.113
Acid density/(kg/m ³)	1160
Acid viscosity/(mPa • s)	0.7

Flow prediction

To facilitate the modification of the hole distribution, we use blind pipe units of uniform length, each unit being 9.3 m long. The pipe is evenly divided into ten sections, with each section containing 29 hole units. At the beginning and end of each section, there is a hole unit with a count of 0, which is considered a joint. Therefore, the total length of the pipe is $30 * 10 * 9.3 = 2790\text{m}$. The diameter of the holes remains unchanged, and the number of holes in each hole unit is limited to 0 or 1. The solutions were obtained according to the following combinations: the first case is a uniform distribution, and the second case is carefully adjusted according to the basic strategy of SL to make the flow in each section as equal as possible, with both having a total of 180 holes. Cases 1 and 2 do not consider formation heterogeneity and treat the reservoir as homogeneous with constant porosity and permeability.

As presented in Table 2, Case 1 employs a uniform hole distribution, with each of the 10 segments containing 18 holes. Case 2 implements the core principle of lift enhancement at the leading edge (SL): a reduced hole density in the inlet section and an increased density toward the outlet section. This distribution represents a modified version rather than a direct application of an arithmetic progression. Preliminary tests indicated that the arithmetic progression configuration alone yielded inadequate SL control performance, necessitating these adjustments. The calculated flow rates through the SL holes are shown in Figures 3 and 4. The two cases compare the flow rates of each section as the basic unit, as shown in Fig. 5. The calculation results indicate that when the hole density remains unchanged, due to the presence of frictional resistance, the acid flow rates through the screens in each section are severely uneven, with the acid flow rate near the starting point being much greater than the flow rate at the endpoint. After adjusting the distribution of the holes, the flow rates in each section are essentially uniform, which reflects the correctness of the SL principle.

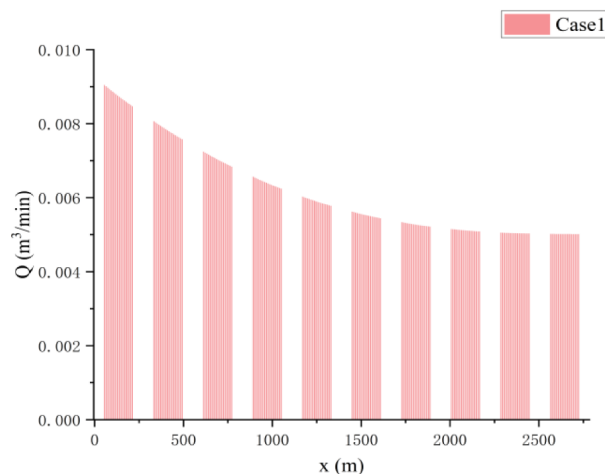


Figure 3—The flow rates of 180 holes for Case1

Table 2—The distribution of holes for cases 1 and 2

Segment number	Number of holes	
	Case1	Case2
1	18	12
2	18	13
3	18	15
4	18	16
5	18	18
6	18	20
7	18	21
8	18	21
9	18	22
10	18	22

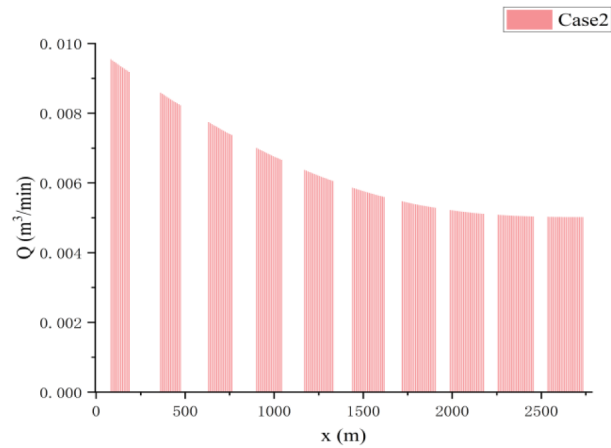


Figure 4—The flow rates of 180 holes for Case2

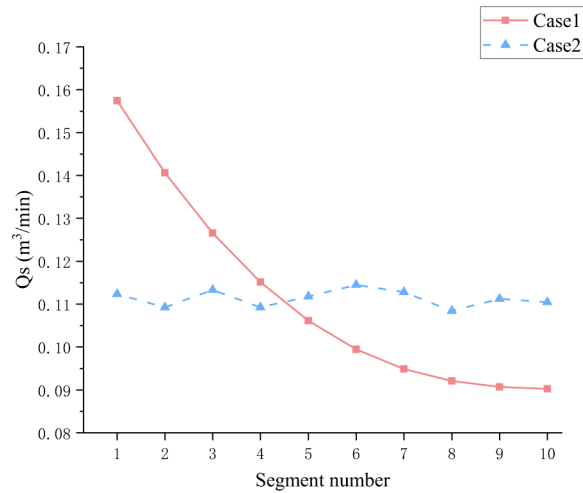


Figure 5—The flow rates of segments for case 1, 2

Wormhole Simulation

After the prediction by the flow prediction program, we have a general understanding of the acid distribution from the SL under two different hole layouts. Next, based on the two-scale model mentioned earlier, we conducted numerical simulations in COMSOL. As mentioned earlier, the liquid flows out from the holes and eventually enters the reservoir. The overall model is shown in the figure below.

The entire reservoir is set to be 50 mm by 20 mm. The cloud map in Figure 6 shows the porosity cloud diagram when the acid solution breaks through the core. In the COMSOL simulation, we did not directly use flow rate as the boundary condition. Instead, we calculated the velocity based on the geometric structure and used it directly as the initial condition. The detailed data is shown below: A 1mm long and 3 mm high rectangular block on the left side, is considered as the hole jet. The right side of the reservoir is regarded as the liquid outlet. Based on the results from the previous section, the jet velocity in the entire simulation is set to five levels: 70,000, 85,000, 100,000, 115,000, and 130,000 cm/min. The porosity of the reservoir is controlled by random variables, simulating scenarios with average porosities of 0.15, 0.13, and 0.11, respectively, corresponding to hypothetical reservoir porosities. During the simulation, the pressure at the inlet is used as a reference. When the pressure drops to 1/100, it is considered that the acid has broken through. The volume of acid at this point is recorded to obtain the PVBT value(Pore Volumes to Breakthrough),defined as the volume of acid divided by the volume of pore when breakthrough occurs. The specific results are as follows:

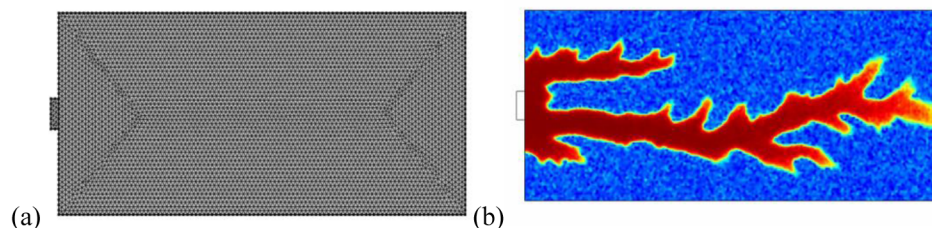


Figure 6—Model mesh in COMSOL (a) and wormhole morphology (porosity cloud diagram) (b)

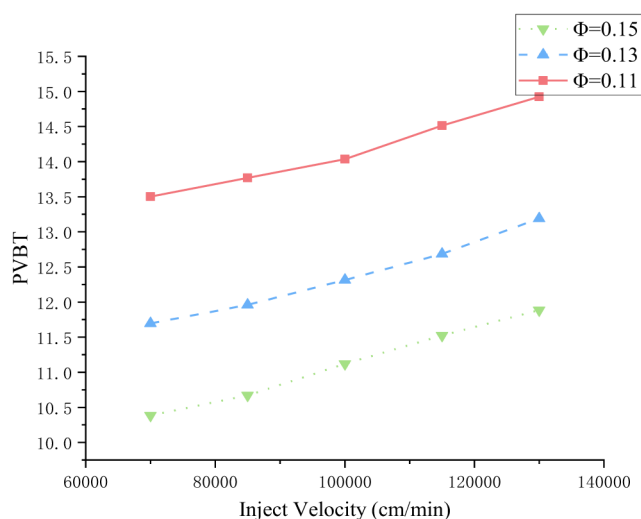


Figure 7—The PVBT of different porosities

We have observed that in the design concept of SL, the spacing between holes is relatively large, about 10 meters. Under an inlet flow rate of 1.113 m³/min, each of the 180 holes can achieve a flow rate of approximately 0.005-0.009 m³/min. Due to the small diameter of the holes (3 mm), the actual velocity of the hole jet is very high, about 70,771-127,389 cm/min. In Panga's paper, five types of wormholes in rock are introduced, which are closely related to the injection velocity. When the injection flow rate is above 60 cc/

min, the wormhole exhibits uniform dissolution. Compared to the optimal injection velocity, the flow rate is already quite high, yet the velocity of the SL holes can easily reach 5000 cc/min, far exceeding the optimal velocity. This indicates that during the operation of SL, a large amount of acid is consumed in reactions, leading to waste. The corresponding PVBT in this case is also much greater than the optimal PVBT, which results in a lower efficiency of wormhole growth at the same acidizing intensity.

M.Buijse et al. proposed a semi-empirical formula, based on which the approximate relationship between PVBT and injection velocity can be obtained as shown in Figure 8 below[10,11]:

$$PVBT = \frac{v^{1/3}}{W_{eff}(1 - \exp(-W_B v^2))^2} \quad (10)$$

with W_{eff} and W_B the semi-empirical coefficient, using the default value.

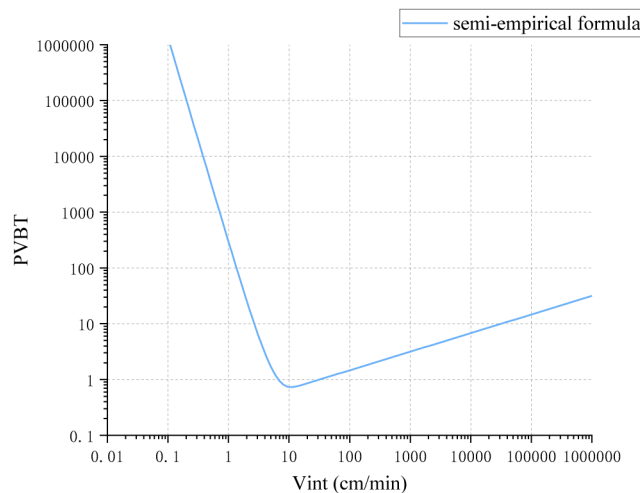


Figure 8—The PVBT and velocity by semi-empirical model

From the Figure 8, we can see that the optimal velocity for hydrochloric acid to react with carbonate rock to form wormholes is around 10 cm/min, but the velocity of the SL hole jet can reach 100,000 cm/min. Corresponding to the results in Figure 7, it can be observed that at the injection velocity of the SL hole, PVBT is approximately 13, while at the optimal injection velocity, PVBT is only around 0.75, making the PVBT almost tens of times the minimum value. The smaller the PVBT, the higher the efficiency of wormhole reservoir by the acid liquid, which leads us to believe that the acid liquid in SL is not acidizing with optimal efficiency.

Hole Configuration Adjustment Based on Wormhole Simulation Results

In the previous text, we demonstrated that by adjusting the distribution of holes in the SL, we could make the discharge of each section almost uniform. However, this did not take into account the heterogeneity of the reservoir and the resulting non-uniformity in the length of the acid-etched wormholes. Now, based on the simulation results, we assume a section of reservoir with non-uniform porosity as shown in the table below and calculate the wormhole length conditions for these two cases. The formation permeability is set to be proportional to porosity

Table 3—The porosities of heterogeneous reservoirs

Segment number	Porosity
1	11%
2	15%
3	15%
4	15%
5	15%
6	13%
7	15%
8	13%
9	11%
10	13%

The wormhole length (r_w) for each segment can be calculated based on PVBT and the flow rate of acid through the holes. The relationship is expressed by Eq (11):

$$r_w = \sqrt{r_{\text{well}}^2 + \frac{Q \times T}{\pi \times \phi \times \text{PVBT}}} \quad (11)$$

with r_w the wormhole length, r_{well} the wellbore diameter, T the acidizing pumping time

Assuming the pumping time is 24 hours, the calculated wormhole lengths for the two cases are shown in the figure below:

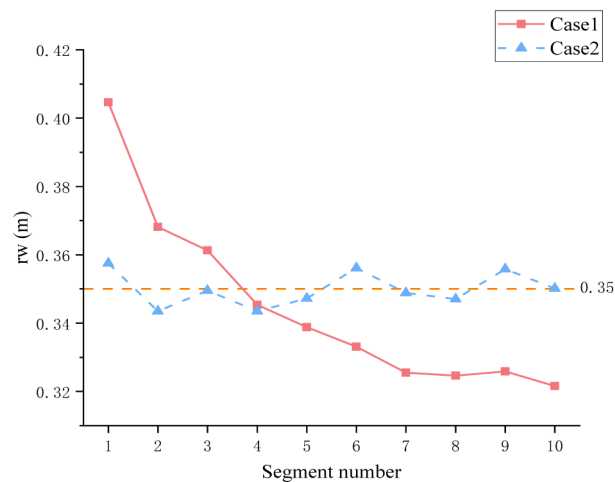


Figure 9—The wormhole lengths for Case 1, 2

Combining this with the flow distribution diagram from the previous text, it can be seen that after considering the impact of reservoir heterogeneity on wormhole length, the acidizing effect further fluctuates due to changes in reservoir porosity. The originally better-performing case 2 shows a significant unevenness in the formed wormholes. Based on this, we further adjust the distribution of the holes to make the wormholes in case 3 more uniform.

As shown in Figures 10 and 11, Case 3 is not as uniform in flow rate as Case 2, but the wormhole length is significantly more uniform. Considering that the adjustment of the holes is based on integer units, Case 3 does not exhibit substantial changes.

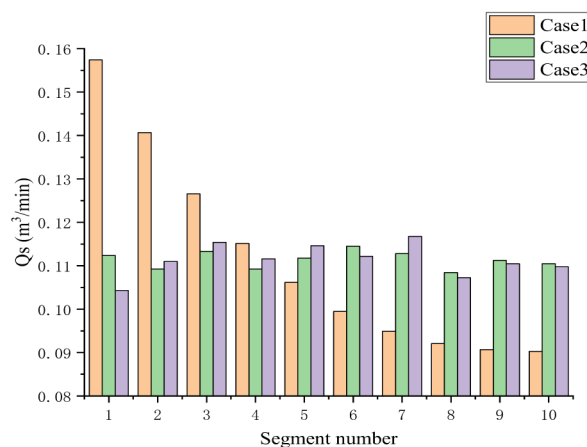


Figure 10—The flow rates of each segment for Case 1, 2, 3

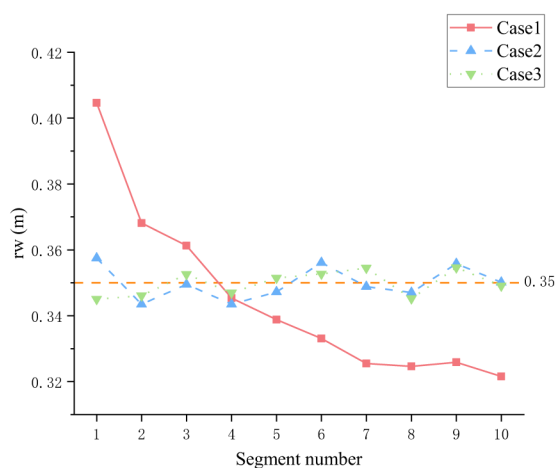


Figure 11—The wormhole lengths of each segment for Cases 1–3

Table 4—The distribution of holes for Cases 1–3

Segment number	Number of holes		
	Case1	Case2	Case3
1	18	12	11
2	18	13	13
3	18	15	15
4	18	16	16
5	18	18	18
6	18	20	19
7	18	21	21
8	18	21	20
9	18	22	21
10	18	22	21

Application

As shown in Figure 12, a pilot application test of the Smart Liner has been conducted in a well of a Middle Eastern oil field. The Smart Liner was installed in the target well in 2021, followed by an intensive acidizing

operation in December of the same year. Production logging tools were used to test the reservoir structure before and after the test, and it was found that the Smart Liner technology significantly improved the uniformity of the production profile structure (with a 90% increase in PI); the liquid production capacity of each reservoir section was significantly enhanced, with the maximum increase approaching 1,007% and the minimum increase being approximately 23%; the differences in liquid production among the various reservoir sections were greatly reduced, with the maximum liquid production difference within the entire section being 341 stb/d, compared to a reduction of 237 stb/d from the maximum liquid production difference before the test (578 stb/d). The liquid production capacity of the oil well was significantly improved, with an increase of 109%.

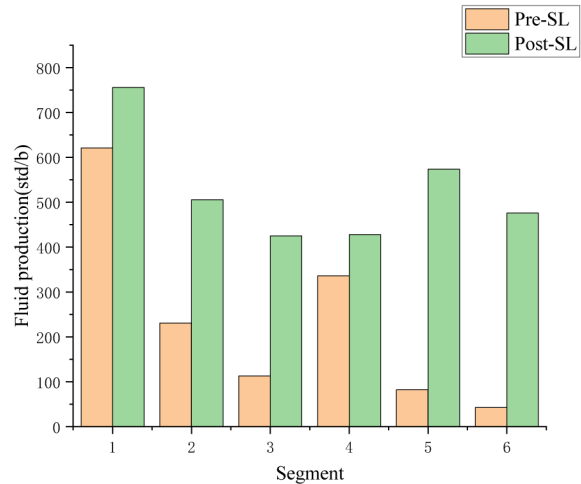


Figure 12—Comparison of fluid production before and after modification of Smart Flow Control Completion in a Middle East oil well

Conclusion

In this paper, we have established a discretized numerical model to calculate the flow distribution of SL, then conducted numerical simulations of acid-etched wormholes based on the two-scale model and the flow rate of SL holes, and finally obtained the wormhole distribution corresponding to uniform distribution and uniform flow cases, and made adjustments accordingly. The relevant conclusions are as follows:

Based on research related to wormholes, we found that under the design concept of SL, the velocity of acid liquid ejected from each hole is very high, far greater than the optimal injection velocity provided by other studies. Considering that the jet may experience velocity attenuation in the annular section—meaning not all acid fluid immediately enters the formation after being ejected from the perforations but instead spreads outward—this conclusion may not be as severe.

When adjusting the SL, the larger the porosity, the easier it is for the acid liquid to breakthrough. Therefore, to maintain uniform wormhole length, the number of holes in reservoirs with large porosity should be reduced, and the number of holes in reservoirs with small porosity should be increased.

Nomenclature

ΔP_f	frictional resistance	Pa
λ	friction factor	1
L	length of the pipe	m

D	diameter of the pipe	m
ρ	density of the acid	kg/m ³
v	velocity of the acid	m/s
f	Fanning friction factor	1
Re	Reynolds number	1
ε	absolute roughness	m
Δp_{hole}	pressure drop through the hole	Pa
q_{hole}	flow rate through the hole	m ³ /s
D_{hole}	diameter of the hole	m
C_D	hole friction factor	1
Q	pipe flow rate	m ³ /s
K	permeability of the rock	mD
μ	viscosity of the acid	mPa · s
∇P	pressure gradient	Pa/m
$R(C_s)$	rate of acid chemical reaction	mol/s
k_c	mass transfer coefficient	m/s
C_s	concentration of acid at the solid-fluid interface	mol/m ³
C_f	cup-mixing concentration of acid in the fluid phase.	mol/m ³
ϕ	porosity	1
t	injection time	s
a_v	interfacial area per unit volume of the medium	1/m
α	dissolving power of acid	1
ρ_s	the rock density.	kg/m ³
r_p	pore radius	m
r_{well}	wellbore diameter	m
T	acidizing pumping time	s

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